

# HPC Evolution

## Hardware

| machine                          | power | processors                                       | memsize/real<br>virtual                                                                        | flops           | OS         | feature                                                             |
|----------------------------------|-------|--------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------|------------|---------------------------------------------------------------------|
| IBM 360                          |       | IBM                                              | Real 8MB<br>Virtual 16MB                                                                       | 3-16 MIPS       | MVS        |                                                                     |
| Control Data                     | 60KW  | 1 at 36MHz                                       | Real 60bit word<br>length<br>64Kwords SCM<br>384 kB<br>Level 2 LCM<br>memory up to 3<br>Mbytes | 10-36<br>MFLOPS |            | pipelined processor                                                 |
| Cray 1                           | 90KW  | 1 at 80 Mhz                                      | 8 Mbytes                                                                                       | 160<br>MFLOPS   | COS        | vector                                                              |
| Cray XMP4                        | 300KW | 4 at 110MHz                                      | 128Mbytes<br>(shared mem)                                                                      | 800<br>MFLOPS   | Unic<br>os | multi-tasking,<br>SSD                                               |
| Cray T90                         | 800KW | 32 at<br>450MHz                                  | 8 Gbytes<br>(shared mem)                                                                       | 57<br>GFLOPS    | Unic<br>os | \$35M !!                                                            |
| Convex                           |       |                                                  |                                                                                                |                 |            |                                                                     |
| Mini-<br>computers               |       |                                                  |                                                                                                |                 |            | a whole plethora of<br>these ... all different<br>processors and OS |
| Digital PDP                      |       |                                                  |                                                                                                |                 |            |                                                                     |
| Digital VAX                      |       |                                                  |                                                                                                |                 |            |                                                                     |
| Data<br>General                  |       |                                                  |                                                                                                |                 |            |                                                                     |
| Prime                            |       |                                                  |                                                                                                |                 |            |                                                                     |
| CACHE AND<br>MICROPRO<br>CESSORS |       |                                                  |                                                                                                |                 |            | Affects everyone                                                    |
| IBM Power                        |       | RS6000<br>range of<br>procs over<br>decades      |                                                                                                |                 |            | Workstation - then<br>clusters                                      |
| Connection<br>Machines<br>CM1    |       | Up to 4096,<br>1bit<br>Later models<br>use SPARC | 4k per<br>processor,<br>512MB max                                                              |                 |            | *LISP<br>Dataflow architecture<br>3-torus interconnect              |
| Sun SPARC                        |       | SPARC range<br>of<br>processors                  |                                                                                                |                 |            | Workstation then<br>clusters                                        |

|                       |              |                                                    |                                            |                         |                |                                                                         |
|-----------------------|--------------|----------------------------------------------------|--------------------------------------------|-------------------------|----------------|-------------------------------------------------------------------------|
|                       |              | over decades                                       |                                            |                         |                |                                                                         |
| Sun E10000            |              | SPARC at 400MHZ up to 64 CPUS                      | 64GBytes shared                            | 32GFLOP S               | Solaris (Unix) | Crossbar interconnect                                                   |
| IBM SP2               | 8KW          | RS6000 processor up to 16 nodes 30-100 MHz         | 16-512 Mbyte per node                      | 13-1500 MFlops per node | AIX            | proprietary interconnect                                                |
| MIPS                  |              | Range of processors over decades                   |                                            |                         |                |                                                                         |
| SGI Origin            | 4KW per rack | MIPS R10000, R12K, R14K 1 to 512 nodes             | 64-512MB per node, ccNUMA, Up to 2TB Total |                         | IRIX           | DSM numalink, hypercube topology                                        |
|                       |              |                                                    |                                            |                         |                |                                                                         |
| Intel 286 PC          |              |                                                    | 32bit no ecc                               |                         |                | DOS!                                                                    |
| Intel 386 PC          |              |                                                    | 32bit no ecc                               |                         |                | Linux                                                                   |
|                       |              |                                                    |                                            |                         |                |                                                                         |
| Beowulf               |              |                                                    |                                            |                         |                | ethernet                                                                |
| MPI                   |              |                                                    |                                            |                         |                | MPI starts to become universal to support programming on multiple nodes |
| Cray T3D/E            |              | DEC Alpha 2176 nodes max 600MHz                    | 64MB - 2GB RAM per node                    | > 1TFLOP                | Unicos/mk      | 6-way router/ 3D torus message passing                                  |
| Cray UV               |              | Intel Xeon processors Up to 2,000 cores            | 16TB ccNUMA up to 16 TB                    |                         | SLES           | DSM Numalink                                                            |
|                       |              |                                                    |                                            |                         |                |                                                                         |
| Intel X86-64 Clusters |              |                                                    | 64bit ECC                                  |                         | Linux          | IB +MPI                                                                 |
| Other interconnects   |              |                                                    |                                            |                         |                | Myrinet,                                                                |
| SGI Altix ICE         |              | Two socket Xeon blades Largest at NASA 6,400 nodes | Cluster                                    | 6 PFLOP                 | SLUES/REHEL    | Blade based cluster multi-rail IB                                       |

|                                     |      |                                                   |                             |                       |       |                                                   |
|-------------------------------------|------|---------------------------------------------------|-----------------------------|-----------------------|-------|---------------------------------------------------|
| Bandwagon everyone makes clusters!! |      |                                                   |                             |                       |       | IB and commodity microprocessors become universal |
| GPU Bandwagon clusters!!            |      |                                                   |                             |                       |       | IB                                                |
| Frontier                            | 25MW | 9,472 AMD Epyc CPU @2GHz + 37,888 AMD MI250X GPUS | 512GB Per CPU 128GB per GPU | 1.3 ExaFLOP Sustained | Linux | Cray Slingshot                                    |

## Software and programming environments

| Decade           | OS                                                                               | Programming env                  | languages                                                         | parallel programming                                          |
|------------------|----------------------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------|
| 1960's           | each vendor has their own<br>IBM OS/360<br>CDC NOS and SCOPE<br>ICL GEORGE       | fixed to hardware                | Assembler<br>COBOL<br>FORTRAN                                     | none                                                          |
| 1970's           | Proprietary vendor ,<br>enhanced for memory paging and virtual memory<br>DEC VMS | fixed to hardware                | Assembler<br>COBOL<br>FORTRAN<br>PASCAL<br>LISP<br>C (K&R)        | tasking on multiprocessor systems                             |
| 1980's           | Proprietary Unix variants<br>IRIX, AIX, Solaris, BSD, AT&T                       | fixed to hardware                | C<br>Fortran                                                      | (multi-) tasking                                              |
| Communications ! |                                                                                  |                                  |                                                                   | TCP/IP and sockets from Unix and Linux standardise networking |
| 1990's           | Unix + Linux                                                                     | Proprietary and Open Source      | C<br>Fortran<br>C++                                               | MPI<br>OpenMP                                                 |
| 2000's           | Linux + unix                                                                     | Open Source plus others eg Intel | Java!?<br>C<br>C++<br>Fortran                                     | MPI<br>(openmp)                                               |
| 2010's           | Linux                                                                            | Open Source plus Intel + NVidia  | C<br>C++<br>Fortran<br>GPUs in use - mostly proprietary like CUDA | MPI<br>OpenMP<br>Hybrid GPU                                   |

|        |       |                                       |                                                                       |                                                                                |
|--------|-------|---------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 2020's | Linux | Open Source<br>plus Intel +<br>NVidia | Fortran<br>Python!<br>C<br>C++<br>CUDA<br>a variety of<br>other niche | Hybrid<br>MPI<br>OpenMP (GPU<br>and CPU)<br>CUDA, other<br>GPU<br>environments |
|        |       |                                       |                                                                       |                                                                                |
|        |       |                                       |                                                                       |                                                                                |
|        |       |                                       |                                                                       |                                                                                |